CMPG 325 Final Technical Report

Project: CMPG325-2026-066
Client: CLI-066 Kagiso Mechanical Engineering
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1. Executive summary

This report presents the final implementation of a three-tier extended-star Cisco campus network for Kagiso Mechanical Engineering. The implementation uses IPv4 VLSM addressing, departmental VLANs, inter-VLAN routing, DHCP relay, OSPF, ACL-based security, Rapid-PVST+ redundancy, port security, DNS, and an internal HTTP/file application server.

The working Packet Tracer project is `CLI066_Mbodi_40779750_M3_FINAL.pkt`. Supporting configuration records and evidence are stored in `Milestone2/`.

2. Requirements and design

The design supports Admin/HR, Engineering Design, Production, Finance/Procurement, ICT Technical, Management/Servers, and Guest WiFi. The topology contains one edge router, one Layer 3 distribution switch, four access switches, servers, end-user devices, printers, and a redundant Engineering uplink.

The assigned IPv4 block is `192.168.35.0/24`. Largest-first VLSM allocation provides Engineering with the largest subnet and separate /28 networks for the remaining VLANs.

3. Addressing plan

VLAN	Purpose	Network	Gateway
10	Admin / HR	`192.168.35.32/28`	`192.168.35.33`
20	Engineering Design	`192.168.35.0/27`	`192.168.35.1`
30	Production Workshop	`192.168.35.48/28`	`192.168.35.49`
40	Finance / Procurement	`192.168.35.96/28`	`192.168.35.97`
50	ICT Technical	`192.168.35.64/28`	`192.168.35.65`
99	Management / Servers	`192.168.35.112/28`	`192.168.35.113`
100	Guest WiFi	`192.168.35.80/28`	`192.168.35.81`
Core link	R-EDGE to DLS1	`192.168.35.252/30`	`.253` / `.254`

4. Implementation summary

DLS1 provides the seven SVIs, DHCP relay to `192.168.35.114`, inter-VLAN routing, OSPF Area 0, Rapid-PVST+ root-primary operation, trunking, and ACL enforcement. R-EDGE provides the routed core link, OSPF, default-route configuration, and NAT/PAT configuration.

Access-switch uplink mapping was verified with CDP:

- ALS-ADM `Gi0/1` to DLS1 `Gig0/2`
- ALS-ENG `Gi0/1` to DLS1 `Fa0/21`
- ALS-ENG `Gi0/2` to DLS1 `Fa0/22`
- ALS-PRD `Gi0/1` to DLS1 `Fa0/23`
- ALS-FIN `Gi0/1` to DLS1 `Fa0/24`

SVR-DC is configured through the Packet Tracer GUI with static address `192.168.35.114/28`, DHCP, and DNS. SVR-FP is configured through the GUI with static address `192.168.35.119/28` and HTTP enabled.

5. Assigned networking feature

The assigned feature is IPv4 VLSM subnetting. It is implemented through the seven non-overlapping VLAN networks, correct SVI masks and gateways, and DHCP scopes matching each VLSM range. DHCP leases were verified for Engineering, Admin, Guest, and

Finance.

The client change request is enforced with seven named inbound ACLs. Engineering, Admin, and ICT are authorised for appropriate server access; Production, Finance, and Guest traffic is restricted according to the CR4 policy.

6. Validation results

Test	Result
VLSM subnet and SVI verification	PASS
Engineering, Admin, Guest, and Finance DHCP leases	PASS
VLAN and trunk verification	PASS
OSPF neighbor and routing verification	PASS
Engineering authorised server access	PASS
Production denied server access	PASS
ACL counters and policy verification	PASS
STP redundant-link failover and restoration	PASS
Port-security violation detection	PASS
DNS and HTTP service verification	PASS
NAT/PAT Internet verification	Not fully testable because Cloud-PT supplied no usable WAN address or next hop

Full screenshots are indexed in `Milestone2/evidence/README.md`.

7. Security and resilience

The design uses VLAN segmentation, named ACLs, management isolation, port security with sticky MAC addresses, PortFast, BPDU Guard, Rapid-PVST+, and a redundant Engineering uplink. The STP test demonstrated that the standby Engineering path forwards after the primary path is disconnected and that the original state can be restored.

8. Limitations and lessons learned

Packet Tracer's Cloud-PT model did not provide a usable Internet DHCP lease or ISP next hop, so NAT/PAT could not be demonstrated end-to-end. Internal routing and all available services were validated instead. The implementation also required explicit dot1q encapsulation on the Catalyst 3560 and careful CDP verification because the physical port labels differed from the initial worksheet.

9. Conclusion

The final Packet Tracer implementation satisfies the internal network, VLSM, service, security, and resilience requirements demonstrated by the evidence pack. The GitHub portfolio contains the project file, technical documentation, PDFs, and organised screenshots. The 15-20 minute INSET video demonstration has been recorded and uploaded.

10. Rubric traceability matrix

Rubric criterion	Weight	Evidence	Result
Client requirements and network design	15%	M1 PDFs (`Milestone1/`), topology screenshots, BOM, cabling schedule	Met
Packet Tracer implementation and functionality	35%	`M3_FINAL.pkt`, DHCP leases, DNS, HTTP, inter-VLAN ping, OSPF, STP	Met
Assigned networking feature (VLSM)	20%	Addressing plan, VLSM arithmetic check, SVI/DHCP verification	Met
GitHub portfolio	15%	README, MD/PDF reports, evidence folder, commit history	Met
Technical demonstration (INSET video)	15%	`project_video.mp4` — recorded and uploaded	Met

11. References

Cisco Systems. (2023). *Catalyst 3560 Series Switches — LAN Base Software Configuration Guide*. Cisco Press.

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Ruehle, R. (2024). *Packet Tracer Lab Manual*. 4th edn. Indianapolis, IN: Cisco Press.

Stevens, R. (2018). *TCP/IP Illustrated*. 2nd edn. Boston, MA: Addison-Wesley.

Tanenbaum, A. (2021). *Computer Networks*. 6th edn. Harlow: Pearson Education.

Westcott, D. and Gerald, D. (2022). *Computer Networking Demystified*. New York, NY: McGraw-Hill.

12. Submission references

- Final Packet Tracer file: `Milestone3/CLI066_Mbodi_40779750_M3_FINAL.pkt`
- Milestone 2 implementation report: `Milestone2/01_Implementation_Report.pdf`
- Milestone 2 configuration worksheet: `Milestone2/02_Device_Configuration_Commands.pdf`
- Evidence index: `Milestone2/evidence/README.md`
- GitHub portfolio: `https://github.com/mulondimbodi/CMPG325-2026-066-CLI066-Mbodi-40779750`